# BEING A LOCAL COMPLETE INTERSECTION MORPHISM IS FPQC LOCAL ON THE BASE FOR ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

*Editorial note.* Statement span is lines 11315-11319 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 11321-11368, the complete author proof (46 body lines).

*Editorial note.* Navigation stubs: 7 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

*Editorial note.* extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

*Editorial note.* Text is retained verbatim, including the author's own typos and placeholders.

*Editorial note.* Counted B side (both complete; spaces more general).

*Allowed standard prerequisites (editorial).*

- Explicit prerequisite boundary. More on Morphisms, lemma descending-property-lci (schemes side); More on Morphisms, lemma lci-syntomic-local-source (card L16201); Morphisms, lemma etale-syntomic; chapter lemma flat-base-change-lci; Topologies on Spaces chart lemmas.

*Source reference.* Original lemma 069I: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 11228–11231. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 069N: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 11368–11368. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 069P: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 11368–11368. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 02GQ: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 11368–11368. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 03MQ: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 11368–11368. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

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*Date*: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

*Source reference.* Original lemma 03MS: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 11368–11368. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 0419: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 11368–11368. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

**Lemma 0.1.** *The property $\mathcal{P}(f) =$"$f$ is a local complete intersection morphism" is fpqc local on the base.*

*Proof.* Let $S$ be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over $S$. Let $\{Y_i \to Y\}$ be an fpqc covering (Topologies on Spaces, Definition 03MQ). Let $f_i : X_i \to Y_i$ be the base change of $f$ by $Y_i \to Y$. If $f$ is a local complete intersection morphism, then each $f_i$ is a local complete intersection morphism by Lemma 069I.

Conversely, assume each $f_i$ is a local complete intersection morphism. We may replace the covering by a refinement (again because flat base change preserves the property of being a local complete intersection morphism). Hence we may assume $Y_i$ is a scheme for each $i$, see Topologies on Spaces, Lemma 0419. Choose a scheme $V$ and a surjective étale morphism $V \to Y$. Choose a scheme $U$ and a surjective étale morphism $U \to V \times_Y X$. We have to show that $U \to V$ is a local complete intersection morphism of schemes. By Topologies on Spaces, Lemma 03MS we have that $\{Y_i \times_Y V \to V\}$ is an fpqc covering of schemes. By the case of schemes (More on Morphisms, Lemma 069N) it suffices to prove the base change

$$U \times_Y Y_i = U \times_V (V \times_Y Y_i) \longrightarrow V$$

of $U \to V$ by $V \times_Y Y_i \to V$ is a local complete intersection morphism. We can write this as the composition

$$U \times_Y Y_i \longrightarrow (V \times_Y X) \times_Y Y_i = V \times_Y X_i \longrightarrow V \times_Y Y_i$$

The first arrow is an étale morphism of schemes (as a base change of $U \to V \times_Y X$) and the second arrow is a local complete intersection morphism of schemes as a flat base change of $f_i$. The result follows as being a local complete intersection morphism is syntomic local on the source and since étale morphisms are syntomic (More on Morphisms, Lemma 069P and Morphisms, Lemma 02GQ). □